

CBSE Grade 9 Physics

Motion Numericals Worksheet

PREMIUM PRACTICE PACK · EQUATIONS OF MOTION FOCUS

CHAPTER

Motion

CLASS

Grade 9 CBSE

MAIN SKILL

Equations of Motion

RECOMMENDED TIME

60 – 75 minutes

TEACHER NOTE

This worksheet upgrades the handwritten questions into a clean classroom-ready practice sheet. The problem set is aligned to CBSE Grade 9 Motion: speed, velocity, acceleration, retardation, free fall, stopping distance, and the three equations of motion.

1. Formula Cheat Sheet

A. Average Speed and Average Velocity

Quantity	Formula	Meaning
Average Speed	$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$	Scalar quantity. Depends on the complete path length.
Average Velocity	$\text{Average Velocity} = \frac{\text{Total Displacement}}{\text{Total Time}}$	Vector quantity. Depends on change in position and direction.

B. Important Shortcut Cases

Case	Formula / Result
Equal distances with speeds v_1 and v_2	$\text{Average Speed} = \frac{2 v_1 v_2}{v_1 + v_2}$
Equal time intervals with speeds v_1 and v_2	$\text{Average Speed} = \frac{v_1 + v_2}{2}$
Return trip to the same point	Average Velocity = 0, because displacement = 0

💡 CORE IDEA

Speed cares about **distance**. Velocity cares about **displacement**. This single difference explains most average speed vs average velocity questions.

C. Three Standard Equations of Motion

EQUATIONS OF UNIFORMLY ACCELERATED MOTION

$$v = u + at$$

1ST EQUATION

$$s = ut + \frac{1}{2} a t^2$$

2ND EQUATION

$$v^2 - u^2 = 2 a s$$

3RD EQUATION

Equation	Use When
$v = u + at$	Time, acceleration, initial velocity, and final velocity are involved.
$s = ut + \frac{1}{2} a t^2$	Distance / displacement is asked and time is given.
$v^2 - u^2 = 2 a s$	Time is not given or not required.

D. Symbols and Meanings

Symbol	Meaning
u	Initial velocity
v	Final velocity
a	Acceleration
t	Time
s	Distance / Displacement

E. Derived / Reduced Formulas

1. Motion from Rest

If the body starts from rest, $u = 0$

$$v = at$$

$$s = \frac{1}{2} at^2$$

$$v^2 = 2as$$

2. Stopping Distance

If the body comes to rest, $v = 0$

$$0 = u^2 + 2as$$

$$s = \frac{u^2}{2|a|}$$

3. Retardation / Deceleration

If speed decreases, acceleration is negative.

$$a = \frac{v - u}{t}$$

4. Free Fall

$a = g$; take $g = 10 \text{ m/s}^2$ for Grade 9

$$v = gt$$

$$s = \frac{1}{2} gt^2$$

$$v^2 = 2gs$$

F. Unit Conversion Box

Conversion	Value
1 km/h	$\frac{5}{18} \text{ m/s}$
1 m/s	$\frac{18}{5} \text{ km/h}$
1 minute	60 s
1 hour	3600 s
1 cm/s ²	0.01 m/s ²

2. Enhanced Practice Questions

Solve all questions using the equations of motion wherever acceleration is involved. For average speed and average velocity, always identify total distance, total displacement, and total time separately.

SECTION A — AVERAGE SPEED AND AVERAGE VELOCITY

QUESTION 1

Speed Battle on a Straight Road

A car moves on a straight road with a speed of 60 m/s for 10 minutes. Then it moves with a speed of 50 m/s for the next 5 minutes.

1. Find the total distance covered.
2. Find the average speed.
3. Find the average velocity, assuming the car keeps moving in the same straight direction.

QUESTION 2

Morning Run Challenge

A boy runs 100 m in 3 minutes and then runs the next 50 m in 1 minute.

1. Find the total distance covered.
2. Find the total time taken.
3. Find the average speed of the boy.

QUESTION 3

Boat Ride and Return Trip

A boat travels from a pier to an island at a speed of 60 km/h for 4 minutes. It then returns to the same pier at a speed of 120 km/h in 2 minutes.

1. Find the distance from the pier to the island.
2. Find the total distance travelled.
3. Find the average speed for the whole journey.
4. Find the average velocity for the whole journey.

SECTION B — ACCELERATION AND RETARDATION

QUESTION 4

Bullet Train Launch

A bullet train starts from rest and reaches a speed of 300 km/h in 6 minutes with uniform acceleration.

1. Find the acceleration of the train.
2. Find the distance covered during this time.

QUESTION 5

Bus Slowing Down

A bus reduces its speed from 60 km/h to 30 km/h in 3 seconds.

1. Find the acceleration of the bus.
2. Find the distance travelled during this time.

QUESTION 6

Playground Slide Motion

A boy starts sliding down from rest on a playground slide with a uniform acceleration of 10 cm/s^2 .

1. Find his velocity after 10 s.
2. Find the distance covered in 10 s.

QUESTION 7

Emergency Braking of a Train

A train is travelling at 100 km/h. Brakes are applied and it undergoes a uniform retardation of 10 m/s^2 .

1. Find the time taken to come to rest.
2. Find the distance travelled before stopping.

QUESTION 8

Cricket Ball Coming to Rest

A cricket ball hit by a batsman moves with a velocity of 150 km/h. It comes to rest after 10 seconds due to friction and air resistance.

1. Find the acceleration of the ball.
2. Find the distance travelled before stopping.

SECTION C — HIGHER-SPEED MOTION

QUESTION 9

Rocket Speed Boost

A rocket moving at 43,200 km/h increases its speed to 86,400 km/h in 2.5 hours.

1. Find the acceleration of the rocket in m/s^2 .
2. Find the distance covered during this speed change.

QUESTION 10**Fast-Moving Body**

A body is moving with an initial velocity of 50 m/s. It accelerates uniformly at 20 m/s^2 for 10 seconds.

1. Find the final velocity.
2. Find the distance covered.
3. Find the average speed during the time interval.

SECTION D — ADDITIONAL CBSE-STYLE QUESTIONS**QUESTION 11****Scooter Acceleration and Cruise**

A scooter starts from rest and accelerates uniformly at 2 m/s^2 for 8 seconds. After that, it moves with the speed gained for the next 12 seconds.

1. Find the speed after 8 seconds.
2. Find the distance covered during acceleration.
3. Find the distance covered during uniform motion.
4. Find the average speed for the entire journey.

QUESTION 12**Elevator Motion**

An elevator starts from rest and moves upward. It reaches a speed of 6 m/s in 3 seconds. Then it slows down uniformly and comes to rest in the next 2 seconds.

1. Find the distance covered while speeding up.
2. Find the distance covered while slowing down.
3. Find the total distance covered.
4. Find the average speed for the whole motion.

QUESTION 13**Free Fall Numeral**

A stone is dropped from rest from a height. Take $g = 10 \text{ m/s}^2$.

1. Find the velocity of the stone after 3 seconds.
2. Find the distance fallen after 3 seconds.

QUESTION 14

Bike Without Time Data

A bike starts from rest and accelerates uniformly at 4 m/s^2 . It covers a distance of 200 m.

1. Find the final velocity.
2. Find the time taken.
3. Find the average speed.

3. Step-by-Step Answer Key and Pedagogy

HOW TO READ THE SOLUTIONS

Each solution follows the same physics habit: write given values, convert units, choose the formula, substitute carefully, and interpret the sign of acceleration or velocity.

Solution 1: Speed Battle on a Straight Road

GIVEN

$$v_1 = 60 \text{ m/s}, \quad t_1 = 10 \text{ min} = 600 \text{ s}$$

$$v_2 = 50 \text{ m/s}, \quad t_2 = 5 \text{ min} = 300 \text{ s}$$

CALCULATION

Distance in first part:

$$s_1 = v_1 \times t_1 = 60 \times 600 = 36,000 \text{ m}$$

Distance in second part:

$$s_2 = v_2 \times t_2 = 50 \times 300 = 15,000 \text{ m}$$

Total distance:

$$s_1 + s_2 = 36,000 + 15,000 = 51,000 \text{ m}$$

Total time:

$$600 + 300 = 900 \text{ s}$$

FORMULA USED

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

SUBSTITUTION & RESULT

$$\text{Average Speed} = \frac{51,000}{900} = 56.67 \text{ m/s}$$

Since the car moves in the same straight direction, displacement equals distance.

$$\text{Average Velocity} = \frac{51,000}{900} = 56.67 \text{ m/s}$$

✓ FINAL ANSWER

Total distance = 51,000 m | Average speed = 56.67 m/s | Average velocity = 56.67 m/s

⚠ COMMON TRAP

Do not directly average 60 and 50 because the time intervals are unequal.

Solution 2: Morning Run Challenge

GIVEN

$$s_1 = 100 \text{ m}, \quad t_1 = 3 \text{ min} = 180 \text{ s}$$

$$s_2 = 50 \text{ m}, \quad t_2 = 1 \text{ min} = 60 \text{ s}$$

CALCULATION

$$\text{Total distance} = 100 + 50 = 150 \text{ m}$$

$$\text{Total time} = 180 + 60 = 240 \text{ s}$$

FORMULA USED

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

SUBSTITUTION & RESULT

$$\text{Average Speed} = \frac{150}{240} = 0.625 \text{ m/s}$$

✓ FINAL ANSWER

Total distance = 150 m | Total time = 240 s | Average speed = 0.625 m/s

⚠ COMMON TRAP

Average speed is not found by averaging individual speeds unless equal time intervals are given.

Solution 3: Boat Ride and Return Trip

GIVEN

First journey: $v_1 = 60 \text{ km/h}$, $t_1 = 4 \text{ min} = \frac{4}{60} \text{ h}$

Return journey: $v_2 = 120 \text{ km/h}$, $t_2 = 2 \text{ min} = \frac{2}{60} \text{ h}$

CALCULATION

Distance from pier to island:

$$s_1 = v_1 \times t_1 = 60 \times \frac{4}{60} = 4 \text{ km}$$

Return distance:

$$s_2 = 120 \times \frac{2}{60} = 4 \text{ km}$$

$$\text{Total distance} = 4 + 4 = 8 \text{ km}$$

$$\text{Total time} = 4 + 2 = 6 \text{ min} = \frac{6}{60} \text{ h} = 0.1 \text{ h}$$

FORMULA USED

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} \quad \text{Average Velocity} = \frac{\text{Total Displacement}}{\text{Total Time}}$$

SUBSTITUTION & RESULT

$$\text{Average Speed} = \frac{8}{0.1} = 80 \text{ km/h}$$

Since the boat returns to the starting point, displacement = 0.

$$\text{Average Velocity} = \frac{0}{0.1} = 0$$

✓ FINAL ANSWER

Distance to island = 4 km | Total distance = 8 km | Average speed = 80 km/h | Average velocity = 0

💡 TIP

Distance is 8 km, but displacement is 0. That is why average speed is not zero, but average velocity is zero.

Solution 4: Bullet Train Launch

GIVEN

$$u = 0, \quad v = 300 \text{ km/h}, \quad t = 6 \text{ min} = 360 \text{ s}$$

UNIT CONVERSION

$$v = 300 \times \frac{5}{18} = 83.33 \text{ m/s}$$

FORMULA USED

$$v = u + at$$

SUBSTITUTION

$$\begin{aligned} 83.33 &= 0 + a \times 360 \\ a &= \frac{83.33}{360} = 0.231 \text{ m/s}^2 \end{aligned}$$

DISTANCE (2ND EQUATION)

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 0.231 \times (360)^2 \\ s &= 15,000 \text{ m} = 15 \text{ km} \end{aligned}$$

✓ FINAL ANSWER

$$a = 0.231 \text{ m/s}^2 \quad | \quad s = 15,000 \text{ m} = 15 \text{ km}$$

💡 TIP

Always convert km/h into m/s before using equations of motion with seconds.

Solution 5: Bus Slowing Down

GIVEN

$$u = 60 \text{ km/h}, \quad v = 30 \text{ km/h}, \quad t = 3 \text{ s}$$

UNIT CONVERSION

$$u = 60 \times \frac{5}{18} = 16.67 \text{ m/s}$$

$$v = 30 \times \frac{5}{18} = 8.33 \text{ m/s}$$

FORMULA USED

$$v = u + at$$

SUBSTITUTION

$$8.33 = 16.67 + a \times 3$$

$$a = \frac{8.33 - 16.67}{3} = \frac{-8.34}{3} = -2.78 \text{ m/s}^2$$

DISTANCE

$$s = \frac{u + v}{2} \times t = \frac{16.67 + 8.33}{2} \times 3 = 12.5 \times 3 = 37.5 \text{ m}$$

✓ FINAL ANSWER

$$a = -2.78 \text{ m/s}^2 \quad | \quad s = 37.5 \text{ m}$$

💡 TIP

Negative acceleration does not always mean the object is moving backward. Here it means the bus is slowing down.

Solution 6: Playground Slide Motion

GIVEN

$$u = 0, \quad a = 10 \text{ cm/s}^2, \quad t = 10 \text{ s}$$

UNIT CONVERSION

$$a = 10 \text{ cm/s}^2 = 0.10 \text{ m/s}^2$$

FORMULA USED & SUBSTITUTION

$$v = u + at = 0 + 0.10 \times 10 = 1 \text{ m/s}$$

$$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 0.10 \times (10)^2 = 0.05 \times 100 = 5 \text{ m}$$

✓ FINAL ANSWER

$$v = 1 \text{ m/s} \quad | \quad s = 5 \text{ m}$$

💡 TIP

10 cm/s^2 looks small, but over time it still increases velocity.

Solution 7: Emergency Braking of a Train

GIVEN

$$u = 100 \text{ km/h}, \quad v = 0, \quad a = -10 \text{ m/s}^2$$

UNIT CONVERSION

$$u = 100 \times \frac{5}{18} = 27.78 \text{ m/s}$$

TIME TO STOP (1ST EQUATION)

$$v = u + at$$

$$0 = 27.78 - 10t$$

$$t = \frac{27.78}{10} = 2.78 \text{ s}$$

STOPPING DISTANCE (3RD EQUATION)

$$v^2 - u^2 = 2as$$

$$0 - (27.78)^2 = 2 \times (-10) \times s$$

$$-771.73 = -20s$$

$$s = \frac{771.73}{20} = 38.6 \text{ m}$$

✓ FINAL ANSWER

$$t = 2.78 \text{ s} \quad | \quad s = 38.6 \text{ m}$$

⚡ SHORTCUT

$$\text{Stopping distance: } s = \frac{u^2}{2|a|}$$

Solution 8: Cricket Ball Coming to Rest

GIVEN

$$u = 150 \text{ km/h}, \quad v = 0, \quad t = 10 \text{ s}$$

UNIT CONVERSION

$$u = 150 \times \frac{5}{18} = 41.67 \text{ m/s}$$

FORMULA USED & SUBSTITUTION

$$v = u + at$$

$$0 = 41.67 + a \times 10$$

$$a = \frac{-41.67}{10} = -4.17 \text{ m/s}^2$$

DISTANCE BEFORE STOPPING

$$s = \frac{u + v}{2} \times t = \frac{41.67 + 0}{2} \times 10 = 20.835 \times 10 = 208.35 \text{ m}$$

✓ FINAL ANSWER

$$a = -4.17 \text{ m/s}^2 \quad | \quad s = 208.35 \text{ m}$$

💡 TIP

When an object stops, final velocity $v = 0$.

Solution 9: Rocket Speed Boost

GIVEN

$$u = 43,200 \text{ km/h}, \quad v = 86,400 \text{ km/h}, \quad t = 2.5 \text{ h}$$

UNIT CONVERSION

$$t = 2.5 \times 3600 = 9,000 \text{ s}$$

$$u = 43,200 \times \frac{5}{18} = 12,000 \text{ m/s}$$

$$v = 86,400 \times \frac{5}{18} = 24,000 \text{ m/s}$$

ACCELERATION

$$a = \frac{v - u}{t} = \frac{24,000 - 12,000}{9,000} = \frac{12,000}{9,000} = 1.33 \text{ m/s}^2$$

DISTANCE

$$s = \frac{u + v}{2} \times t = \frac{12,000 + 24,000}{2} \times 9,000 = 18,000 \times 9,000 = 162,000,000 \text{ m} = 162,000 \text{ km}$$

✓ FINAL ANSWER

$$a = 1.33 \text{ m/s}^2 \quad | \quad s = 162,000,000 \text{ m} = 162,000 \text{ km}$$

💡 TIP

Large speed does not always mean large acceleration. Acceleration depends on change in velocity per unit time.

Solution 10: Fast-Moving Body

GIVEN

$$u = 50 \text{ m/s}, \quad a = 20 \text{ m/s}^2, \quad t = 10 \text{ s}$$

FINAL VELOCITY (1ST EQUATION)

$$v = u + at = 50 + 20 \times 10 = 250 \text{ m/s}$$

DISTANCE (2ND EQUATION)

$$s = ut + \frac{1}{2}at^2 = 50 \times 10 + \frac{1}{2} \times 20 \times (10)^2$$
$$s = 500 + 1,000 = 1,500 \text{ m}$$

AVERAGE SPEED

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{1,500}{10} = 150 \text{ m/s}$$

⚡ SHORTCUT

For uniformly accelerated motion: $\text{Average Speed} = \frac{u + v}{2} = \frac{50 + 250}{2} = 150 \text{ m/s}$

✅ FINAL ANSWER

$v = 250 \text{ m/s}$		$s = 1,500 \text{ m}$		Average speed = 150 m/s
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Solution 11: Scooter Acceleration and Cruise

GIVEN — PHASE 1 (ACCELERATION)

$$u = 0, \quad a = 2 \text{ m/s}^2, \quad t = 8 \text{ s}$$

SPEED AFTER 8 S

$$v = u + at = 0 + 2 \times 8 = 16 \text{ m/s}$$

DISTANCE DURING ACCELERATION

$$s_1 = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 2 \times (8)^2 = 64 \text{ m}$$

PHASE 2 (UNIFORM MOTION)

The scooter moves uniformly at 16 m/s for 12 s.

$$s_2 = v \times t = 16 \times 12 = 192 \text{ m}$$

AVERAGE SPEED

$$\text{Total distance} = 64 + 192 = 256 \text{ m}$$

$$\text{Total time} = 8 + 12 = 20 \text{ s}$$

$$\text{Average speed} = \frac{256}{20} = 12.8 \text{ m/s}$$

✓ FINAL ANSWER

Speed after 8 s = 16 m/s | $s_1 = 64 \text{ m}$ | $s_2 = 192 \text{ m}$ | Average speed = 12.8 m/s

💡 TIP

This is a two-phase problem. Solve one phase at a time.

Solution 12: Elevator Motion

PHASE 1 — SPEEDING UP

$$u = 0, \quad v = 6 \text{ m/s}, \quad t = 3 \text{ s}$$

$$\begin{aligned}\text{Average speed} &= \frac{u + v}{2} = \frac{0 + 6}{2} = 3 \text{ m/s} \\ s_1 &= 3 \times 3 = 9 \text{ m}\end{aligned}$$

PHASE 2 — SLOWING DOWN

$$u = 6 \text{ m/s}, \quad v = 0, \quad t = 2 \text{ s}$$

$$\begin{aligned}\text{Average speed} &= \frac{6 + 0}{2} = 3 \text{ m/s} \\ s_2 &= 3 \times 2 = 6 \text{ m}\end{aligned}$$

TOTAL

$$\text{Total distance} = 9 + 6 = 15 \text{ m}$$

$$\text{Total time} = 3 + 2 = 5 \text{ s}$$

$$\text{Average speed} = \frac{15}{5} = 3 \text{ m/s}$$

✓ FINAL ANSWER

Distance speeding up = 9 m | Distance slowing down = 6 m | Total = 15 m | Average speed = 3 m/s

Solution 13: Free Fall Numeral

GIVEN

$$u = 0, \quad g = 10 \text{ m/s}^2, \quad t = 3 \text{ s}$$

VELOCITY AFTER 3 S

$$v = u + gt = 0 + 10 \times 3 = 30 \text{ m/s (downward)}$$

DISTANCE FALLEN

$$s = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2} \times 10 \times (3)^2 = 5 \times 9 = 45 \text{ m}$$

✓ FINAL ANSWER

$$v = 30 \text{ m/s (downward)} \quad | \quad s = 45 \text{ m}$$

💡 TIP

For free fall, acceleration is due to gravity.

Solution 14: Bike Without Time Data

GIVEN

$$u = 0, \quad a = 4 \text{ m/s}^2, \quad s = 200 \text{ m}$$

FORMULA USED (3RD EQUATION — NO TIME GIVEN)

$$v^2 - u^2 = 2 a s$$

$$v^2 - 0 = 2 \times 4 \times 200 = 1,600$$

$$v = 40 \text{ m/s}$$

FINDING TIME (1ST EQUATION)

$$v = u + at$$

$$40 = 0 + 4t \implies t = 10 \text{ s}$$

AVERAGE SPEED

$$\text{Average Speed} = \frac{u + v}{2} = \frac{0 + 40}{2} = 20 \text{ m/s}$$

✓ FINAL ANSWER

$$v = 40 \text{ m/s} \quad | \quad t = 10 \text{ s} \quad | \quad \text{Average speed} = 20 \text{ m/s}$$

💡 TIP

When acceleration is uniform, Average Speed = $\frac{u + v}{2}$.

4. No-Time Alternative Method Section

In these modified scenarios, time is not directly given. The aim is to make students identify whether the situation has **equal distance intervals** or **equal time intervals**, then choose the correct average speed formula.

🔗 BIG DECISION RULE

Equal distance? Use $\frac{2 v_1 v_2}{v_1 + v_2}$. **Equal time?** Use $\frac{v_1 + v_2}{2}$. **Return trip?** Average velocity = 0 because displacement = 0.

Case 1: Equal Distance Intervals — Straight-Line Motion with Non-Zero Displacement

QNT1**Half Journey Slow, Half Journey Fast**

An object covers the first half of a straight-line distance with speed v and the second half with speed $2v$.

1. Find the average speed.

Solution NT1

For equal distances, do not take the simple average of speeds.

FORMULA USED

$$\text{Average Speed} = \frac{2 v_1 v_2}{v_1 + v_2}$$

SUBSTITUTION

Here, $v_1 = v$ and $v_2 = 2v$.

$$\text{Average Speed} = \frac{2 \times v \times 2v}{v + 2v} = \frac{4v^2}{3v} = \frac{4v}{3}$$

✓ FINAL ANSWER

$$\text{Average speed} = \frac{4v}{3}$$

⚠ COMMON TRAP

Do not write $\frac{v + 2v}{2} = \frac{3v}{2}$. That is only for equal time, not equal distance.

Case 2: Equal Distance Return Trip — Displacement Becomes Zero**QNT2****Boat Goes and Comes Back**

A boat travels from a pier to an island at 60 km/h and returns to the same pier at 120 km/h. The distance in both directions is the same.

1. Find the average speed.
2. Find the average velocity.

Solution NT2

AVERAGE SPEED (EQUAL DISTANCES)

$$\text{Average Speed} = \frac{2 v_1 v_2}{v_1 + v_2} = \frac{2 \times 60 \times 120}{60 + 120} = \frac{14,400}{180} = 80 \text{ km/h}$$

AVERAGE VELOCITY

Since the boat returns to the starting point, displacement = 0.

$$\text{Average Velocity} = \frac{\text{Total Displacement}}{\text{Total Time}} = 0$$

✓ FINAL ANSWER

Average speed = 80 km/h | Average velocity = 0

💡 TIP

Same starting and ending point means zero displacement, not zero distance.

Case 3: Equal Time Intervals

QNT3

Car Moves at Two Speeds for Equal Times

A car moves at 60 m/s for one time interval and then at 50 m/s for the same time interval in the same direction.

1. Find the average speed.

Solution NT3

For equal time intervals, use the simple average of speeds.

$$\text{Average Speed} = \frac{v_1 + v_2}{2} = \frac{60 + 50}{2} = 55 \text{ m/s}$$

✓ FINAL ANSWER

Average speed = 55 m/s

💡 TIP

Equal time means arithmetic mean of speeds.

Case 4: Equal Time but Opposite Directions

QNT4

Runner Goes East and Then West

A runner moves east with a speed of 8 m/s for some time. Then he moves west with a speed of 6 m/s for the same time.

1. Find the average speed.
2. Find the average velocity.

Solution NT4

Let the time in each part be t .

AVERAGE SPEED

Distance east = $8t$. Distance west = $6t$.

$$\text{Total distance} = 8t + 6t = 14t$$

$$\text{Total time} = 2t$$

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{14t}{2t} = 7 \text{ m/s}$$

AVERAGE VELOCITY

Taking east as positive:

$$\text{Displacement} = 8t - 6t = 2t \text{ (east)}$$

$$\text{Average Velocity} = \frac{\text{Displacement}}{\text{Total Time}} = \frac{2t}{2t} = 1 \text{ m/s (east)}$$

✓ FINAL ANSWER

Average speed = 7 m/s | Average velocity = 1 m/s (east)

⚠ COMMON TRAP

Average speed uses **total distance**. Average velocity uses **displacement**.

Case 5: Equal Distance with Real Numbers

QNT5**City Drive**

A car covers the first half of a straight road at 40 km/h and the second half at 60 km/h.

1. Find the average speed.

Solution NT5

The two distance intervals are equal, so use the equal-distance formula.

$$\text{Average Speed} = \frac{2 v_1 v_2}{v_1 + v_2} = \frac{2 \times 40 \times 60}{40 + 60} = \frac{4,800}{100} = 48 \text{ km/h}$$

✓ FINAL ANSWER

Average speed = 48 km/h

⚠ COMMON TRAP

Many students write 50 km/h, but that is wrong because distance intervals are equal, not time intervals.

Case 6: Equal Time with Real Numbers**QNT6****Two Equal Time Speed Runs**

A scooter moves at 30 km/h for one time interval and 50 km/h for the same time interval.

1. Find the average speed.

Solution NT6

The two time intervals are equal, so use the simple average of speeds.

$$\text{Average Speed} = \frac{v_1 + v_2}{2} = \frac{30 + 50}{2} = 40 \text{ km/h}$$

✓ FINAL ANSWER

Average speed = 40 km/h

5. Final Revision Box

Situation	Correct Formula / Result	Memory Hook
Equal distance	$\text{Avg Speed} = \frac{2 v_1 v_2}{v_1 + v_2}$	Distance equal → harmonic-type average
Equal time	$\text{Avg Speed} = \frac{v_1 + v_2}{2}$	Time equal → simple average
Return to starting point	Average Velocity = 0	Displacement becomes zero
Stopping motion	$v = 0$ and acceleration is negative	Retardation means slowing down
Time missing in equations of motion	Use $v^2 - u^2 = 2 a s$	No time? Use third equation

6. Student Challenge

The 5-Second Rule



Before solving any Motion numerical, **pause for 5 seconds** and identify: Is it average speed? Is it acceleration? Is time missing? Is the object stopping? **This choice of formula is half the solution.**